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/**

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* @file           : main.c
* @author          : Auto-generated by STM32CubeIDE
* @brief           : Task 1 : three IRQ with same priority
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*/

#include <stdint.h>
#include <stdio.h>

#define IRQNO_TIM6 54
#define IRQNO_UART3 39
#define IRQNO_OTG_FS 77

#define NVIC_IPR_BASE ((uint32_t*)0xE000E400)
#define NVIC_ISER_BASE ((uint32_t*)0xE000E100)
#define NVIC_ISPR_BASE ((uint32_t*)0xE000E200)

// Function to configure priority (8-bit per IRQ, top 4 bits used)
void configure_priority_for_irq(uint8_t irq_no, uint8_t
priority_value){
    uint8_t iprx = irq_no / 4;        // Each IPR register holds 4 IRQs
    uint8_t pos  = (irq_no % 4) * 8; // Each IRQ uses 8 bits

    uint32_t *ipr = NVIC_IPR_BASE + iprx;

    *ipr &= ~(0xFF << pos);           // Clear old priority
    *ipr |= ((priority_value & 0xFF) << pos); // Set new priority
}

// Function to enable IRQ
void enable_irq(uint8_t irq_no){
    uint8_t reg_index = irq_no / 32;
    uint8_t bit_pos   = irq_no % 32;

    NVIC_ISER_BASE[reg_index] |= (1 << bit_pos);
}

// Function to set pending IRQ (simulate IRQ trigger)
void set_pending_irq(uint8_t irq_no){
    uint8_t reg_index = irq_no / 32;
    uint8_t bit_pos   = irq_no % 32;

    NVIC_ISPR_BASE[reg_index] |= (1 << bit_pos);
}

// ----- ISR -----
void TIM6_DAC_IRQHandler(void){
    printf("TIM6 ISR executed\n");
}

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void USART3_IRQHandler(void){
    printf("USART3 ISR executed\n");
}

void OTG_FS_IRQHandler(void){
    printf("OTG_FS ISR executed\n");
}

// ----- MAIN -----
int main(void)
{
    // 1. Configure all interrupts with SAME priority
    // In Cortex-M4, lower numeric value = higher actual priority
    // We use 0x80 (middle priority) for all
    configure_priority_for_irq(IRQNO_TIM6, 0x80);
    configure_priority_for_irq(IRQNO_UART3, 0x80);
    configure_priority_for_irq(IRQNO_OTG_FS, 0x80);

    // 2. Enable all IRQs
    enable_irq(IRQNO_TIM6);
    enable_irq(IRQNO_UART3);
    enable_irq(IRQNO_OTG_FS);

    // 3. Trigger all interrupts simultaneously (simulate)
    set_pending_irq(IRQNO_TIM6);
    set_pending_irq(IRQNO_UART3);
    set_pending_irq(IRQNO_OTG_FS);

    printf("NVIC configured. Pending IRQs set.\n");

    // In real STM32, NVIC will call ISR automatically
    while(1); // Wait for interrupts
}

```